

Do I/O Schedulers Perform Better in Virtual Machines or in Docker Containers?

The I/O schedulers in Linux manage the order in which the I/O requests of processes are handled by the disk subsystem. This article compares the performance of different I/O schedulers in Linux virtual machines and Docker containers, offering insights into optimising disk access for improved system performance.



Process schedulers are essential to the functionality and responsiveness of Linux systems and are well-known to the majority of users. Their primary role is to manage the distribution of system resources among processes, including the CPU, memory, and I/O hardware. The process scheduler's objective is to maximise system throughput, reduce response time, and ensure fairness among all active processes.

In addition to process schedulers, Linux also has I/O schedulers that manage the order in which processes' I/O requests are handled by the disk subsystem. The I/O scheduler performs a similar role to the process scheduler, but instead manages the processes' disk access rather than the CPU. In situations where multiple I/O requests

from different processes compete for disk access, the I/O scheduler is crucial to the system's overall performance.

Despite the use of flash and solid-state storage, disk access remains the most expensive form of data access, being slower than data access from RAM. I/O schedulers optimise disk access requests to improve performance.

This article compares the efficiency of different I/O schedulers when SSDs are used, and the results are particularly relevant for processes that involve intensive disk access. The comparison is made between running benchmarks on Ubuntu 20.04 and using Docker and containers (where files must first be copied into the container to run the benchmarks).

"Prior to Ubuntu 19.04 with Linux 5.0 or Ubuntu 18.04.3 with Linux

4.15, the multiqueue I/O scheduling was not enabled by default and just the deadline, cfq and noop I/O schedulers were available by default. For Ubuntu 19.10 with Linux 5.0 or Ubuntu 18.04.3 with Linux 5.0 onwards, multiqueue is enabled by default providing the bfq, kyber, mq-deadline and none I/O schedulers. For Ubuntu 19.10 with Linux 5.3 the deadline, cfq and noop I/O schedulers are deprecated." (*Ubuntu Wiki*, 2019)

The above statement can be taken as true for Ubuntu 20.04 too. Hence, for the comparison only the following I/O schedulers were taken into account:

- none I/O scheduler
- mq-deadline I/O scheduler
- bfq I/O scheduler
- kyber I/O scheduler

I/O scheduler commands

To check the available I/O schedulers, run the following command:

```
$ cat /sys/block/sda/queue/scheduler
```

The output would be something like:

```
[mq-deadline] none
```

This means that the currently selected I/O scheduler is *mq-deadline*, and other options are available in the I/O scheduler, for example, *none*.